Question	Answer	Marks	Partial Marks
7(c)(ii)	64	**2**	**FT** *their* **(c)(i)** correctly evaluated provided at least 2 terms but not the original equation **M1** for substituting $x = 4$ into *their* **(c)(i)**
7(c)(iii)	(0, 6) $\left(\frac{4}{3}, \frac{98}{27}\right)$ oe	**4**	**M1** for *their* derivative = 0 or $\frac{dy}{dx} = 0$ soi **B1** for $x = 0$ and $x = \frac{4}{3}$ **M1dep** for substituting one of *their x* values into $y = 2x^3 - 4x^2 + 6$ soi
8(a)(i)	3 22 43 48 50	**2**	**B1** for 4 correct or **M1** for one error in adding.
8(a)(ii)	correct diagram	**3**	**B1FT** *their* **(a)(i)** for 5 correct heights **B1** for 5 points at upper ends of intervals on correct vertical line **B1FT dep on at least B1** for increasing curve through *their 5* points After 0 scored, **SC1** for 4 of *their* points correctly plotted
8(a)(iii)	35 to 38	**1**	
8(b)	Correct box-and-whisker diagram 1.45 1.57 1.64 1.71 1.83	**4**	**B1** for median 1.64 drawn **B1** for LQ 1.57 drawn **B1** for UQ 1.71 drawn If 0 scored **SC1** for 1.64, 1,57 or 1.71 seen
9(a)	1350 or 1354….	**6**	**M2** for $20^2 - 13^2$ or **M1** for $BC^2 + 13^2 = 20^2$ **A1** for $\sqrt{231}$ or 15.2 or 15.19 to 15.20 **M1** for 20×24 and 13×24 and *their* 15.2×24 **M1** for [½ ×] *their* 15.2×13
9(b)	2370 or 2369 to 2371… cao	**1**	
9(c)	24.6 or 24.58 to 24.59	**4**	**M3** for sin [...] = $\frac{13}{\sqrt{20^2 + 24^2}}$ oe or **M2** for $\sqrt{20^2 + 24^2}$ or $\sqrt{24^2 + 20^2 - 13^2}$ or **M1** for $AF^2 = 20^2 + 24^2$ or $24^2 + 20^2 - 13^2$ or **M1** for correct angle identified
10(a)	0.75 3 7 3 0.75	**3**	**B2** for 4 or 3 correct or **B1** for 2 correct

Question	Answer	Marks	Partial Marks
10(b)	correct curve	**4**	**B3FT** for 8 or 9 correct plots **B2FT** for 6 or 7 correct plots **B1FT** for 4or 5 correct plots
10(c)	Accept any integer ≥ 8	**1**	
10(d)	line $y = 4 - \frac{1}{2}x$ ruled	**B3**	**B2** for $[y =]4 - \frac{1}{2}x$ identified or **B1** for ruled line with gradient $-\frac{1}{2}$ or **B1** for ruled line through (0, 4) but not $y = 4$
	0.2 to 0.3 4.2 to 4.3	**B1**	
11(a)	20	**2**	**M1** for $\frac{360}{18}$ or $180 - \frac{16 \times 180}{18}$
11(b)	4.5	**2**	**M1** for $\frac{BE}{6.75} = \frac{5.2}{5.2 + 2.6}$ oe
11(c)	5.8[0] or 5.798 to 5.799	**3**	**M2** for $2 \times \sqrt[3]{\frac{780}{32}}$ oe or **M1** for $\sqrt[3]{\frac{780}{32}}$ or $\sqrt[3]{\frac{32}{780}}$ or $\frac{2^3}{l^3} = \frac{32}{780}$
11(d)	$QN = NR$ [given]	**B1**	
	Two correct pairs of angles with reasons from angle PQN = angle SRN alternate angle QPN = angle RSN alternate angle PNQ = angle SNR [vertically] opposite	**B2**	**B1** for any correct pair of angles with reason or two correct pairs of angles with no/wrong reasons
	ASA [implies congruent]	**B1**	dep on B1 B2